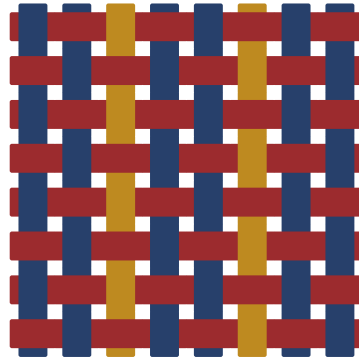




Concentration of Measure

Proof notes, organized by engine

北极甜虾(剑心犹在！)

June 28, 2026

dyed in indigo, madder & weld; ruled in iron-gall. woven on LOOM.

OVERVIEW · THE EIGHT ENGINES

□□□□

In one line: a function of many *weakly dependent* variables is nearly constant. Every concentration inequality is one **engine** for proving this; they differ only in the structure they consume: independence, bounded differences, convexity, or curvature and isoperimetry. Learn the engine before the formula.

Trigger. You want a small-probability bound on $f(X_1, \dots, X_n)$ deviating far from its mean/median.

Ask yourself one question: *which structure do I have?* Then reach for the engine below.

Cheat-sheet: which engine to use

Engine	Structure it eats	Main tools / section
Cramér–Chernoff	independence + MGF	Laplace transform, optimize λ , rate function (§1)
Sub-Gaussian / sub-exp.	<i>shape</i> of the tail	ψ -Orlicz norms, Bernstein (§2)
Bounded diff. / martingale	change one coordinate	Azuma, McDiarmid (§3)
Variance / entropy	tensorization	Efron–Stein, Herbst, log-Sobolev (§4)
Geometry / isoperimetry	metric <i>and</i> measure	sphere/Gaussian isoperimetry, Borell–TIS (§5)
Talagrand convex dist.	convexity + product space	convex-distance inequality (§6)
Matrix concentration	spectrum + trace exponential	Lieb, matrix Bernstein (§7)
Suprema / chaining	metric entropy	Dudley, generic chaining (§8)

Reading guide: §1–§2 are the trunk (the continuation of Appendix A); read them until fluent. §3–§4 push “sums” to “any bounded-difference function”. §5–§6 are the geometric heart that gives “measure concentration” its name. §7–§8 are two exits toward ML/TCS.

CONTENTS

Overview · the eight engines	1
1 The Cramér–Chernoff engine	2
2 Sub-Gaussian and sub-exponential	3
3 Bounded differences and martingales	5
4 The variance and entropy methods	6
5 Geometry and isoperimetry	8

6	Talagrand's convex distance	9
7	Matrix concentration	10
8	Suprema and chaining	12
	Open threads · where this goes next	13

1 THE CRAMÉR–CHERNOFF ENGINE

□□□□ The trunk. Every later engine just works on one step of this pipeline.

In one line: to bound the probability that a random quantity strays far from its mean by something **exponentially small**, exponentiate the tail through the Laplace transform (the MGF) and pick a good λ to push the right-hand side down. The whole subject is variations on this single move.

The paradigm

Trigger. You want an exponentially small bound on $\Pr[X \geq a]$, with the MGF of X under control.

For every $\lambda > 0$,

$$\Pr[X \geq a] = \Pr[e^{\lambda X} \geq e^{\lambda a}] \leq \mathbb{E}[e^{\lambda X}] e^{-\lambda a} \quad (\text{Markov on } e^{\lambda X}),$$

then take the infimum over λ . Writing the cumulant generating function $\psi_X(\lambda) = \log \mathbb{E}[e^{\lambda X}]$,

$$\log \Pr[X \geq a] \leq -\psi_X^*(a), \quad \psi_X^*(a) = \sup_{\lambda > 0} (\lambda a - \psi_X(\lambda)),$$

where ψ_X^* is the **Legendre transform** (the rate function) of ψ_X .

Three-line heuristic: (1) independence $\Rightarrow$ MGFs *multiply*, which is the *only* reason a “sum” gets an exponential bound; (2) *center* first, $X \leftarrow X - \mathbb{E}X$, for the cleanest argument; (3) bound each $\mathbb{E}[e^{\lambda X_i}]$ by one *termwise convex* estimate. Every theorem in the subject just swaps that estimate and that λ .

Cheat-sheet: the three opening moves

Tool	Setting	Tail bound / when to use
Markov	$Y \geq 0$	$\Pr[Y > \alpha \mathbb{E}Y] < 1/\alpha$; opening move, first-order info only
Chebyshev	$\text{Var } X < \infty$	$\Pr[X - \mathbb{E}X > t] \leq \text{Var } X/t^2$; second-order info
Chernoff (master)	MGF exists	$\Pr[X \geq a] \leq e^{-\psi_X^*(a)}$; want an exponential tail

Main results & proof skeletons

Theorem 1.1 (*Chernoff master bound*). If the MGF of X is finite near 0, then $\Pr[X \geq a] \leq e^{-\psi_X^*(a)}$.

Proof. Skeleton: (i) Markov on $e^{\lambda X}$; (ii) take logs to get $\log \Pr[X \geq a] \leq \psi_X(\lambda) - \lambda a$; (iii) take the infimum over $\lambda > 0$. [fill in: add convexity of ψ_X , the attaining λ , and the equality case] ■

Lemma 1.2 (*Additivity: independence makes ψ add*). $X_1, \dots, X_n$ independent $\Rightarrow \psi_{\sum_i X_i} = \sum_i \psi_{X_i}$.

Proof. [fill in: one line: $\mathbb{E}[e^{\lambda \sum X_i}] = \prod \mathbb{E}[e^{\lambda X_i}]$, take logs] ■

Example 1.3 (*The sub-Gaussian seed: a symmetric ± 1 sum*). Let ε_i be independent uniform ± 1 and $S_n = \sum_{i \leq n} \varepsilon_i$. From $\cosh \lambda \leq e^{\lambda^2/2}$ we get $\psi_{S_n}(\lambda) \leq \lambda^2 n/2$, and $\lambda = a/n$ gives

$$\Pr[S_n \geq a] \leq e^{-a^2/2n}, \quad \Pr[|S_n| \geq a] \leq 2e^{-a^2/2n}.$$

[fill in: prove $\cosh \lambda \leq e^{\lambda^2/2}$ by termwise Taylor comparison; this is the seed inequality of the whole subject]

Why it goes this way: $\cosh \lambda \leq e^{\lambda^2/2}$ is the *seed inequality*. Replace it by a different termwise estimate (Hoeffding's lemma $\rightarrow$ bounded variables, Bennett/Bernstein $\rightarrow$ variance-aware) and you grow §2–§4. Swap the estimate, swap λ , that is all.

Tightness: Chernoff gets the large-deviation rate right

Theorem 1.4 (*Cramér, matching lower bound*). Under mild conditions $\lim_{n \rightarrow \infty} \frac{1}{n} \log \Pr\left[\frac{1}{n} \sum_i X_i \geq a\right] = -\psi_{X_1}^*(a)$.

Proof. Skeleton: the upper bound is Thm 1.1+Lem 1.2; the lower bound uses *exponential tilting* (change of measure) to make a the mean under the new measure, then CLT. [fill in: construct the tilted measure $dQ \propto e^{\lambda x} dP$]

[mgf $\leftrightarrow$] This thread is upgraded in §2 to “sub-Gaussian $\equiv$ MGF dominated by $e^{\sigma^2 \lambda^2/2}$ ”.

2 SUB-GAUSSIAN AND SUB-EXPONENTIAL

□□□□ Turn “what does the tail look like” into a norm. Once classified, tail bounds come for free.

In one line: rather than recompute MGFs each time, classify random variables by the *shape of their tail*. Sub-Gaussian = tail no heavier than a Gaussian's; sub-exponential = no heavier than an exponential's. Each family gets an Orlicz norm, and concentration becomes arithmetic on those norms.

The paradigm

Trigger. You want to handle a whole class of tails at once, and have the tail of sums and products follow *mechanically*.

Four equivalent characterizations (sub-Gaussian, write $\|X\|_{\psi_2}$): tail $\Pr[|X| > t] \leq 2e^{-t^2/K^2} \Leftrightarrow$ moments $\|X\|_p \leq K\sqrt{p} \Leftrightarrow$ MGF $\mathbb{E}e^{\lambda X} \leq e^{K^2 \lambda^2/2}$ (after centering) $\Leftrightarrow \mathbb{E}e^{X^2/K^2} \leq 2$.

Three-line heuristic: (1) **a centered bounded variable is sub-Gaussian** (Hoeffding's lemma), so the entire “bounded-difference” family lands in range here; (2) sums of sub-Gaussians stay sub-Gaussian and the proxy variances add: $\|\sum X_i\|_{\psi_2}^2 \leq \sum \|X_i\|_{\psi_2}^2$; (3) a product of two sub-Gaussians is sub-exponential, which is exactly why covariance / quadratic-form estimates drop to the sub-exponential family.

Cheat-sheet: the two tail families

Theorem	Setting	Tail bound / when to use
Hoeffding lemma	$\mathbb{E}X = 0, X \in [a, b]$	$\mathbb{E}e^{\lambda X} \leq e^{\lambda^2(b-a)^2/8}$; bridge bounded $\rightarrow$ sub-Gaussian
Hoeffding ineq.	$X_i \in [a_i, b_i]$ independent	$\Pr[S - \mathbb{E}S \geq t] \leq 2e^{-2t^2 / \sum (b_i - a_i)^2}$
General Hoeffding	X_i sub-Gaussian	$\Pr[\sum a_i X_i \geq t] \leq 2e^{-ct^2 / \ a\ _2^2 K^2}$
Bernstein	X_i sub-exp., $\mathbb{E}X_i = 0$	$\Pr[S \geq t] \leq 2 \exp\left[-c \min\left(\frac{t^2}{\sum v_i^2}, \frac{t}{\max b_i}\right)\right]$
Bennett	$ X_i \leq b$, variance σ^2	variance-aware; Gaussian for small t , Poisson for large t

Main results & proof skeletons

■ **Definition 2.1** (Sub-Gaussian norm). $\|X\|_{\psi_2} = \inf\{K > 0 : \mathbb{E} \exp(X^2/K^2) \leq 2\}$. Call X **sub-Gaussian** if this is finite. (Sub-exponential $\|X\|_{\psi_1}$ replaces X^2 by $|X|$.)

■ **Proposition 2.2** (Four equivalences). For centered X the four characterizations above are pairwise equivalent, with constants differing only by absolute factors.

Proof. Skeleton: tail $\Rightarrow$ moments (integrate the tail) $\Rightarrow$ MGF (sum the series $\sum K^p p^{p/2} \lambda^p / p!$) $\Rightarrow$ tail (Chernoff, §1) $\Rightarrow$ Orlicz. [fill in: track the constants through the four steps; draw it as a cycle] ■

■ **Lemma 2.3** (Hoeffding's lemma). $\mathbb{E}X = 0, X \in [a, b]$ a.s. $\Rightarrow \mathbb{E}e^{\lambda X} \leq e^{\lambda^2(b-a)^2/8}$.

Proof. Skeleton: set $\psi(\lambda) = \log \mathbb{E}e^{\lambda X}$; compute $\psi(0) = \psi'(0) = 0$ and $\psi''(\lambda) = \text{Var}_\lambda(X) \leq (b-a)^2/4$ (the variance under the tilted measure is at most a quarter of the range squared); Taylor remainder. [fill in: prove $\psi'' \leq (b-a)^2/4$ via the variance-range estimate] ■

■ **Theorem 2.4** (Hoeffding's inequality). $X_i \in [a_i, b_i]$ independent, $S = \sum X_i$. Then $\Pr[|S - \mathbb{E}S| \geq t] \leq 2 \exp\left(-\frac{2t^2}{\sum_i (b_i - a_i)^2}\right)$.

Proof. [fill in: feed Lem 2.3 into the master bound of §1 + Lem 1.2, optimize λ] ■

■ **Theorem 2.5** (Bernstein's inequality). X_i independent, centered, sub-exponential. Then the tail is Gaussian for small t and exponential for large t (see the cheat-sheet row).

Proof. Skeleton: the single-term sub-exponential MGF bound $\mathbb{E}e^{\lambda X_i} \leq e^{v_i^2 \lambda^2/2}$ holds only for $|\lambda| < 1/b_i$, so the choice of λ is constrained by $1/b$, producing the two-regime min. [fill in: do the constrained λ -optimization (the boundary between the two regimes)] ■

Why it goes this way: a Gaussian tail comes from "MGF dominated everywhere by $e^{c\lambda^2}$ "; an exponential tail comes from "dominated only for $|\lambda| < 1/b$ ". The constrained λ -

optimization automatically yields Bernstein's $\min(t^2/v, t/b)$ split, which is the whole secret of being *variance-aware*.

Example 2.6 (*Johnson – Lindenstrauss, preview*). The squared length of a Gaussian random projection $\frac{1}{\sqrt{m}}Gx$ is a χ^2 -type sum (sub-exponential), and Bernstein gives $\Pr[|\|\Pi x\|^2 - \|x\|^2| > \epsilon \|x\|^2] \leq 2e^{-cm\epsilon^2}$. [fill in: add the union bound giving $m = O(\epsilon^{-2} \log N)$]

[mgf $\leftrightarrow$] The sub-Gaussian thread is reused throughout §3 (McDiarmid) and §7 (matrix Bernstein) via [subgaussian $\leftrightarrow$].

~~~~~  
 $\rightarrow$  subgaussian  
 Check the constant: does Hoeffding's  $2t^2 / \sum (b-a)^2$  agree with Appendix A's A.1.18 (range  $\leq 1$  gives  $e^{-2a^2/n}$ )?

### 3 BOUNDED DIFFERENCES AND MARTINGALES

□□□□ Upgrade “sum of independents” to “any function that changes little when one coordinate moves”.

In one line: it need not be a sum. As long as  $f(x_1, \dots, x_n)$  changes by a bounded amount when a single coordinate is altered, reveal its randomness *one coordinate at a time* (the Doob martingale); each step has bounded difference, feed it to Hoeffding's lemma from §2, and concentration appears.

#### The paradigm

**Trigger.**  $f$  is not a sum, but has **bounded differences**: altering coordinate  $i$  changes it by  $|f - f'| \leq c_i$ .

Build the **Doob martingale**  $M_k = \mathbb{E}[f \mid X_1, \dots, X_k]$ , so  $M_0 = \mathbb{E}f$ ,  $M_n = f$ , with increments  $|M_k - M_{k-1}| \leq c_k$ . [ subgaussian  $\leftrightarrow$  ] Apply Hoeffding's lemma per step and sum.

Three-line heuristic: (1) **any** function is its own Doob martingale (revealing one coordinate = one step); (2) concentration depends only on *how far each step can jump*, not on how complex  $f$  is; (3) the martingale constant ( $\sum c_i^2$ ) is looser than the entropy method (§4) by a factor, but the proof is shortest and the assumptions weakest.

#### Cheat-sheet: the bounded-difference family

| Theorem         | Setting                                | Tail bound / when to use                                                                      |
|-----------------|----------------------------------------|-----------------------------------------------------------------------------------------------|
| Azuma–Hoeffding | increments $ M_k - M_{k-1}  \leq c_k$  | $\Pr[ M_n - M_0  \geq t] \leq 2e^{-t^2/2 \sum c_k^2}$                                         |
| McDiarmid       | $f$ bounded diff. $c_i$ , $X_i$ indep. | $\Pr[ f - \mathbb{E}f  \geq t] \leq 2e^{-t^2/\sum c_i^2}$ ;<br>workhorse of ML generalization |
| Self-bounding   | $f$ self-bounding                      | one-sided Bernstein type,<br>variance-aware (cf. §4)                                          |
| Freedman        | martingale + predictable var. $V$      | variance-aware martingale tail, tighter when $V$ small                                        |

#### Main results & proof skeletons

**Theorem 3.1** (Azuma–Hoeffding).  $(M_k)$  a martingale,  $|M_k - M_{k-1}| \leq c_k$  a.s. Then  $\Pr[M_n - M_0 \geq t] \leq \exp(-t^2/2 \sum_k c_k^2)$ .

*Proof.* Skeleton: for the increments  $D_k = M_k - M_{k-1}$  we have  $\mathbb{E}[D_k | \mathcal{F}_{k-1}] = 0$  and  $|D_k| \leq c_k$ , so the conditional Hoeffding lemma gives  $\mathbb{E}[e^{\lambda D_k} | \mathcal{F}_{k-1}] \leq e^{\lambda^2 c_k^2/2}$ ; peel off layer by layer via the tower property to get  $\mathbb{E} e^{\lambda(M_n - M_0)} \leq e^{\lambda^2 \sum c_k^2/2}$ ; then Chernoff. [fill in: fill the conditional Hoeffding lemma and the tower-property unrolling]

**Theorem 3.2** (McDiarmid’s bounded-differences inequality).  $X_1, \dots, X_n$  independent,  $f$  with bounded differences  $c_i$ . Then  $\Pr[|f - \mathbb{E}f| \geq t] \leq 2 \exp(-2t^2 / \sum_i c_i^2)$ .

*Proof.* [fill in: feed the Doob martingale  $M_k = \mathbb{E}[f | X_{1:k}]$  into Thm 3.1; independence keeps each increment’s range  $\leq c_k$  (sup minus inf)]

**Example 3.3** (Concentration of the generalization gap).  $f = \sup_{h \in \mathcal{H}} (\mathbb{E}_{\mathcal{D}} \ell_h - \hat{\mathbb{E}} \ell_h)$  has bounded differences  $\leq 1/n$  in each sample, so McDiarmid gives  $\Pr[f - \mathbb{E}f \geq t] \leq e^{-2nt^2}$ . Thus the generalization gap concentrates uniformly around its expectation (= the Rademacher complexity). [fill in: verify the bounded-difference constant  $1/n$ ]

Why it goes this way: McDiarmid splits “uniform convergence” into two things, *concentration* (this section,  $f$  hugs  $\mathbb{E}f$ ) and *the size of the expectation* ( $\mathbb{E}f$  = Rademacher complexity, estimated by the chaining of §8). That is the source of the standard two-step recipe in statistical learning theory.

The truly tight bound comes from the *tensorization* of §4; this section is its “martingale approximation”. [tensorize  $\leftrightarrow$ ]

## 4 THE VARIANCE AND ENTROPY METHODS

□□□□ Tensorization: variance and entropy are subadditive over product measures, so high-dim = one-dim subadditivity + a single-coordinate estimate.

In one line: both variance (Efron–Stein) and entropy (log-Sobolev / Herbst) are *subadditive* over product measures. **Tensorize** a high-dimensional concentration into  $n$  single-coordinate contributions, each needing only a one-dimensional estimate. This route gives the tightest, *variance-aware*, *dimension-free* bounds, and is the analytic engine behind the geometry of §5 and Talagrand (§6).

### The paradigm

**Trigger.** You want the tightest *variance-aware* bound, or to prove a function sub-Gaussian without a “sum / martingale” structure.

**Herbst’s argument:** if the entropy obeys  $\text{Ent}[e^{\lambda f}] \leq \frac{\lambda^2 \sigma^2}{2} \mathbb{E}[e^{\lambda f}]$  for all  $\lambda$ , then  $f - \mathbb{E}f$  is  $\sigma$ -sub-Gaussian. Combine with the **tensorization** of entropy  $\text{Ent}[g] \leq \sum_i \mathbb{E} \text{Ent}_i[g]$  to reduce the problem to a single coordinate.

Three-line heuristic: (1) **Efron–Stein** is the tensorization of variance, giving the *free* (no i.i.d. needed) bound  $\text{Var } f \leq \frac{1}{2} \sum_i \mathbb{E}(f - f^{(i)})^2$ ; (2) replace Var by Ent and “square” by “exponential”: tensorization + a one-dimensional log-Sobolev = exponential concentration (Herbst); (3) the Gaussian measure satisfies a *dimension-free* log-Sobolev inequality (Gross), the algebraic source of the Gaussian concentration in §5.

### Cheat-sheet: variance and entropy

| Theorem              | Setting                                                                                | Gives / when to use                                                                                       |
|----------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Efron–Stein          | product measure, $f^{(i)}$ = resample coord. $i$                                       | $\text{Var } f \leq \frac{1}{2} \sum_i \mathbb{E}(f - f^{(i)})^2$ ; the “bounded difference” for variance |
| Bounded diff. via ES | bounded diff. $c_i$                                                                    | $\text{Var } f \leq \frac{1}{4} \sum c_i^2$ ; variance from the martingale family                         |
| Herbst argument      | $\text{Ent}[e^{\lambda f}] \leq \frac{\lambda^2 \sigma^2}{2} \mathbb{E} e^{\lambda f}$ | $\Rightarrow f$ is $\sigma$ -sub-Gaussian                                                                 |
| Gaussian LSI         | $\gamma = N(0, I_n)$                                                                   | $\text{Ent}[g^2] \leq 2\mathbb{E}\ \nabla g\ ^2$ ; dimension-free                                         |
| Modified LSI         | bounded / Bernoulli coords                                                             | tensorize + single-coord LSI $\Rightarrow$ concentration                                                  |

### Main results & proof skeletons

**Theorem 4.1** (*Efron–Stein inequality*).  $X_1, \dots, X_n$  independent,  $X'_i$  an i.i.d. copy,  $f^{(i)}$  replacing coordinate  $i$  by  $X'_i$ . Then  $\text{Var } f \leq \frac{1}{2} \sum_{i=1}^n \mathbb{E}[(f - f^{(i)})^2]$ .

*Proof.* Skeleton: use the orthogonal martingale-difference decomposition  $\text{Var } f = \sum_i \mathbb{E}[\Delta_i^2]$  with  $\Delta_i = \mathbb{E}[f \mid X_{1:i}] - \mathbb{E}[f \mid X_{1:i-1}]$ ; then Jensen to replace  $\Delta_i$  by the symmetrized  $(f - f^{(i)})$ . [ fill in: fill the orthogonal decomposition + symmetrization, two steps ]

**Theorem 4.2** (*Herbst’s argument*). If  $\text{Ent}[e^{\lambda f}] \leq \frac{\lambda^2 \sigma^2}{2} \mathbb{E}[e^{\lambda f}]$  for all  $\lambda$ , then  $\mathbb{E}[e^{\lambda(f - \mathbb{E}f)}] \leq e^{\lambda^2 \sigma^2 / 2}$ , so  $f$  is  $\sigma$ -sub-Gaussian.

*Proof.* Skeleton: let  $H(\lambda) = \frac{1}{\lambda} \log \mathbb{E} e^{\lambda f}$ ; the entropy inequality is exactly the rewriting  $H'(\lambda) \leq \sigma^2 / 2$ ; integrate to get  $H(\lambda) - H(0^+) \leq \lambda \sigma^2 / 2$ , with  $H(0^+) = \mathbb{E}f$ . [ fill in: fill the differential identity  $\text{Ent}[e^{\lambda f}] = \lambda^2 \mathbb{E} e^{\lambda f} H'(\lambda)$  ]

**Theorem 4.3** (*Gaussian log-Sobolev, Gross*). For  $\gamma = N(0, I_n)$  and smooth  $g$ ,  $\text{Ent}_\gamma[g^2] \leq 2 \mathbb{E}_\gamma \|\nabla g\|^2$ .

*Proof.* [ fill in: fill: the one-dimensional LSI (Ornstein–Uhlenbeck semigroup, or CLT limit from two-point Bernoulli) + tensorization to  $n$  dimensions ]

Why it goes this way: Gaussian LSI + Herbst  $\Rightarrow$  “Gaussian concentration  $e^{-t^2/2L^2}$  for an  $L$ -Lipschitz function, dimension-free”. This is the algebraic path that *computes* the geomet-

ric statement of §5: curvature (the LSI constant) turns directly into the tail rate.

[ tensorize  $\leftrightarrow$  ] Tensorization reappears in §6 in Marton's transportation form.

## ■ 5 GEOMETRY AND ISOPERIMETRY

□□□□ Where "concentration of measure" gets its name: on a metric measure space, a Lipschitz function is almost constant.

In one line: a viewpoint with no MGF at all. On a metric measure space, every 1-Lipschitz function **hugs its median**; this is equivalent to an **isoperimetric** phenomenon, a set of measure  $\geq \frac{1}{2}$ , blown up by  $t$ , swallows  $1 - e^{-t^2}$  of the measure. High-dimensional spheres and Gaussian space behave exactly so; this is the geometric root of "a random vector's length is almost determined" and "a random projection preserves distance".

### The paradigm

**Trigger.**  $f$  is an  $L$ -Lipschitz function on a space with **concentration function**  $\alpha(t)$ .

Then  $\Pr[|f - \text{Med } f| > t] \leq 2\alpha(t/L)$ . Three pillar spaces:

sphere  $S^{n-1}$  :  $\alpha(t) \leq e^{-(n-1)t^2/2}$ ;

Gaussian  $\gamma_n$  :  $\alpha(t) \leq e^{-t^2/2}$  (**dimension-free**);

Hamming cube :  $\alpha(t) \leq e^{-2t^2/n}$ .

Three-line heuristic: (1) concentration  $\equiv$  isoperimetry: the tail bound is the complement of "the  $t$ -blow-up of a half-measure set"; (2) median and mean are interchangeable (differ by  $O(L)$ , same order); (3) the Gaussian case being *dimension-free* is the key miracle, it makes infinite-/high-dimensional processes (§8) possible.

### Cheat-sheet: geometric concentration

| Theorem          | Space                  | Statement / when to use                                                                 |
|------------------|------------------------|-----------------------------------------------------------------------------------------|
| Lévy's lemma     | sphere $S^{n-1}$       | 1-Lip $f$ : $\Pr[ f - \text{Med } f  > t] \leq 2e^{-(n-1)t^2/2}$                        |
| Gaussian conc.   | $\gamma_n = N(0, I_n)$ | $L$ -Lip $f$ : $\Pr[ f - \mathbb{E}f  > t] \leq 2e^{-t^2/2L^2}$ , <i>dimension-free</i> |
| Borell-TIS       | Gaussian process       | $\sup_t X_t$ concentrates around its supremum, variance = $\sup_t \text{Var } X_t$      |
| Gaussian isoper. | $\gamma_n$             | half-spaces are extremal; exact blow-up in $\Phi^{-1}$ form                             |

### Main results & proof skeletons

**Theorem 5.1** (Lévy's lemma / spherical concentration).  $f : S^{n-1} \rightarrow \mathbb{R}$  is 1-Lipschitz. Then  $\Pr[|f - \text{Med } f| > t] \leq 2e^{-(n-1)t^2/2}$ .

*Proof.* Skeleton: spherical **isoperimetry** (Lévy-Schmidt: geodesic balls are extremal)  $\Rightarrow$  for a half-measure set  $A$ , the  $t$ -neighborhood satisfies  $\sigma(A_t) \geq 1 - e^{-(n-1)t^2/2}$ ; take  $A = \{f \leq \text{Med } f\}$ , and Lipschitz pushes  $\{f > \text{Med } f + t\}$  outside  $A_t$ . [ fill in: fill the standard isoperimetry  $\Rightarrow$  concentration translation; cite spherical isoperimetry (do not reprove) ] ■

~~~~~  
Four faces:
concentration $\Leftrightarrow$
isoperimetry $\Leftrightarrow$
log-Sobolev $\Leftrightarrow$
transportation
(Marton).
Herbst is the
"LSI $\Rightarrow$
concentration"
face; the other
three are to be
filled, see §5
and §9 open
threads.

Theorem 5.2 (Gaussian concentration / Borell–TIS). $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -Lipschitz, $X \sim N(0, I_n)$. Then $\Pr[|f(X) - \mathbb{E}f| > t] \leq 2e^{-t^2/2L^2}$, with constants independent of n .

Proof. Either path: (a) Gaussian log-Sobolev (Thm 4.3) + Herbst (Thm 4.2), with smooth approximation of f ; or (b) Gaussian isoperimetry (half-space extremality) gives the blow-up directly. [fill in: take (a): fill the Lipschitz-gradient equivalence $\|\nabla f\| \leq L$ a.e. + smoothing]

Example 5.3 (A random vector's length is almost determined). $x \mapsto \|x\|$ is 1-Lipschitz $\Rightarrow$ for $X \sim N(0, I_n)$, $\|X\|$ concentrates near $\sqrt{n}$ with fluctuation $O(1)$ (independent of n). This is the one-line proof of the “thin-shell” phenomenon. [fill in: fill $\mathbb{E}\|X\| = \sqrt{n}(1 - o(1))$ and substitute]

Why it goes this way: the MGF method (§1) answers “sums”; the geometric method answers “any Lipschitz function”, dimension-free in the Gaussian case, at the cost of needing a “hard” input like isoperimetry / log-Sobolev. The two paths meet at Gaussian concentration: the LSI of §4 is precisely the algebraic proof of Borell–TIS.

Borell–TIS (concentration of the supremum of a Gaussian process) is the “concentration half” of the chaining in §8, chaining controls the *expectation*, this theorem the *fluctuation*. [borelltis $\leftrightarrow$]

~~~~~  
Median vs mean: they differ by  $O(L)$ , but exact isoperimetry is most natural with the median. When must they be distinguished? (heavy tails, asymmetric  $f$ .)

## 6 TALAGRAND'S CONVEX DISTANCE

□□□□ A product space has no geometry, only convexity. Talagrand's convex distance adapts to “how many bad coordinates”.

In one line: on a pure *product* probability space (no sphere-like geometry), Talagrand's **convex distance** yields dimension-free, very strong concentration for convex Lipschitz functions. Its edge over bounded differences (§3): it does not look at the “largest single-coordinate jump” but at “how many coordinates you must change to escape a set”, adapting to sparse dependence.

### The paradigm

**Trigger.**  $f$  on a product space is **convex** Lipschitz, or escaping an event  $A$  requires changing many coordinates.

Define the **convex distance**  $d_T(x, A) = \sup_{\|\alpha\|_2 \leq 1} \inf_{y \in A} \sum_{i: x_i \neq y_i} \alpha_i$  (weighted Hamming). Talagrand:  $\mathbb{E}[e^{d_T(X, A)^2/4}] \leq 1/\Pr[A]$ , hence

$$\Pr[A] \cdot \Pr[d_T(X, A) \geq t] \leq e^{-t^2/4}.$$

Three-line heuristic: (1)  $d_T$  is a min-max of *weighted* Hamming, automatically finding the “cheapest escape direction”; (2) once  $\Pr[A] \geq \frac{1}{2}$ , the convex distance from  $A$  concentrates sub-Gaussianly; (3) for a convex  $L$ -Lipschitz  $f$ , take  $A = \{f \leq \text{Med } f\}$  to get  $\Pr[|f - \text{Med } f| > t] \leq 4e^{-t^2/4L^2}$ , dimension-free.

### Cheat-sheet: convex distance and its corollaries

| Result                   | Setting                            | Statement / when to use                                        |
|--------------------------|------------------------------------|----------------------------------------------------------------|
| Convex-distance ineq.    | product space, any $A$             | $\Pr[A] \Pr[d_T \geq t] \leq e^{-t^2/4}$ ; the master switch   |
| Convex Lipschitz         | $f$ convex, $L$ -Lip on $[0, 1]^n$ | $\Pr[ f - \text{Med } f  > t] \leq 4e^{-t^2/4L^2}$             |
| LIS / combinatorial opt. | $f =$ longest incr. subseq., etc.  | $\sqrt{}$ -scale fluctuation, beats bounded diff.'s $n$ -scale |
| Largest singular value   | random matrix                      | $\sigma_{\max}$ concentrates around its median                 |

## Main results & proof skeletons

**Theorem 6.1** (Talagrand's convex-distance inequality).  $X = (X_1, \dots, X_n)$  a product measure,  $A$  measurable. Then  $\mathbb{E}[e^{d_T(X,A)^2/4}] \leq 1/\Pr[A]$ .

*Proof.* Skeleton (induction on  $n$ , the hardest step in the book): (i) write  $d_T$  via convex duality (the distance to the convex hull  $U_A(x) = \text{conv}\{\mathbf{1}[x_i \neq y_i] : y \in A\}$ ); (ii) condition on the last coordinate and apply the inductive hypothesis to the two cases "in  $A$  / projection in  $A$ "; (iii) a one-dimensional optimization  $\inf_{0 \leq r \leq 1} r \dots$  merges the two cases. [ fill in: fill the three-step induction; the one-dim optimization lemma  $\inf_r e^{(1-r)^2/4}(\cdot)^r$  is the crux ]

**Corollary 6.2** (Concentration of convex Lipschitz functions).  $f : [0, 1]^n \rightarrow \mathbb{R}$  convex and  $L$ -Lipschitz (Euclidean),  $X$  a product measure. Then  $\Pr[|f - \text{Med } f| > t] \leq 4e^{-t^2/4L^2}$ .

*Proof.* [ fill in: use convexity to control  $f(x) - \text{Med } f$  by  $L \cdot d_T(x, \{f \leq \text{Med } f\})$  (convex  $\Rightarrow$  supporting hyperplane), then apply Thm 6.1 ]

**Example 6.3** (Longest increasing subsequence (LIS)). For a random permutation,  $L_n \sim 2\sqrt{n}$ ; changing one value moves it by at most 1, but "certifying  $L_n \geq k$  needs only  $k$  positions", so the convex distance gives fluctuation  $O(n^{1/4})$ , far beating bounded differences'  $O(\sqrt{n})$ . [ fill in: fill the certifiable / self-certifying structure  $\Rightarrow d_T$  bound ]

Why it goes this way: bounded differences only knows "each coordinate has  $c_i$ "; Talagrand knows "how many coordinates must conspire to change the conclusion". When dependence is sparse (few coordinates determine  $f$ ), convex distance replaces  $\sqrt{\sum c_i^2}$  by  $\sqrt{\# \text{critical coords}}$ , the source of the  $n \rightarrow \sqrt{n} \rightarrow n^{1/4}$  chain of improvements in combinatorial optimization.

[ tensorize  $\leftrightarrow$  ] Convex distance can also be seen as the scalar precursor of the spectral concentration of §7.

## 7 MATRIX CONCENTRATION

□□□□ Lift scalar Chernoff/Bernstein to the spectral norm of a random matrix. The obstacle is  $e^{A+B} \neq e^A e^B$ .

In one line: to bound the largest eigenvalue of a sum of independent random matrices, copying §1 stalls because *matrix exponentials do not multiply*. The rescue is **Lieb's concavity** + the trace exponential: use  $\mathbb{E} \operatorname{tr} \exp(\cdot)$  as a surrogate; it is *subadditive*, so the scalar proof transfers almost verbatim, with the dimension entering only at the end as  $\log d$ .

## The paradigm

**Trigger.**  $\sum_i X_i$  is a sum of independent random Hermitian matrices and you want a tail for  $\|\sum_i X_i\|$  (spectral norm).

**Matrix Laplace transform:** for  $\theta > 0$ ,

$$\Pr[\lambda_{\max}(\sum_i X_i) \geq t] \leq e^{-\theta t} \mathbb{E} \operatorname{tr} \exp\left(\sum_i \log \mathbb{E} e^{\theta X_i}\right),$$

where the  $\operatorname{tr} \exp(\sum \dots)$  comes from Lieb's theorem ( $A \mapsto \operatorname{tr} \exp(H + \log A)$  is concave) + Jensen.

Three-line heuristic: (1) turn " $\lambda_{\max} \geq t$ " into  $\operatorname{tr} e^{\theta \sum X_i} \geq e^{\theta t}$  then Markov; (2) **Lieb** lets us push  $\mathbb{E}$  inside  $\operatorname{tr} \exp$  and makes the matrix cumulants *subadditive* (replacing the scalar "MGFs multiply"); (3) the dimension  $d$  enters only through  $\operatorname{tr} \leq d \cdot \lambda_{\max}$ , so it is nearly dimension-free at *low intrinsic dimension*.

## Cheat-sheet: matrix tail bounds

| Theorem          | Setting                                                               | Tail bound / when to use                                                                |
|------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Matrix Laplace   | independent Hermitian $X_i$                                           | the display above; mother of all matrix tails                                           |
| Matrix Chernoff  | $0 \leq X_i \leq RI$                                                  | $\lambda_{\min} / \lambda_{\max}(\sum X_i)$ around $\mu_{\min} / \mu_{\max}$ ; PSD sums |
| Matrix Bernstein | $\mathbb{E} X_i = 0, \ X_i\  \leq L, \ \sum \mathbb{E} X_i^2\  = \nu$ | $\Pr[\ \sum X_i\  \geq t] \leq 2d e^{-t^2/2(\nu + Lt/3)}$                               |
| Intrinsic-dim.   | as above, $\text{intdim} = \nu / \ \cdot\ $                           | $d \rightarrow \text{intrinsic dim}$ , drops the ambient-dimension log                  |

## Main results & proof skeletons

**Theorem 7.1** (*Matrix Laplace transform method, Tropp*). For independent random Hermitian  $X_i$  and  $\theta > 0$ , the matrix Laplace inequality above holds.

*Proof.* Skeleton: (i)  $\Pr[\lambda_{\max} \geq t] \leq e^{-\theta t} \mathbb{E} \operatorname{tr} e^{\theta \sum X_i}$  (Markov +  $\lambda_{\max} \leq \operatorname{tr}$  on PSD); (ii) **Lieb's theorem**  $\Rightarrow A \mapsto \operatorname{tr} \exp(H + \log A)$  concave  $\Rightarrow$  Jensen pushes  $\mathbb{E}$  inside; (iii) peel off term by term to get the subadditive cumulant  $\operatorname{tr} \exp(\sum_i \log \mathbb{E} e^{\theta X_i})$ . [fill in: fill the Lieb  $\Rightarrow$  subadditivity induction; cite Lieb (do not reprove)]

**Theorem 7.2** (*Matrix Bernstein*).  $X_i$  independent,  $\mathbb{E} X_i = 0, \|X_i\| \leq L$ , variance proxy  $\nu = \|\sum_i \mathbb{E} X_i^2\|$ . Then  $\Pr[\|\sum_i X_i\| \geq t] \leq 2d \exp\left(\frac{-t^2/2}{\nu + Lt/3}\right)$ .

*Proof.* [fill in: feed the single-term matrix cgf bound  $\log \mathbb{E} e^{\theta X_i} \leq \frac{\theta^2/2}{1-L\theta/3} \mathbb{E} X_i^2$  into Thm 7.1, optimize  $\theta$ ]

**Example 7.3** (Covariance estimation). For the sample covariance  $\hat{\Sigma}$  of  $n$  independent samples, write  $\hat{\Sigma} - \Sigma = \frac{1}{n} \sum_i X_i$  (centered outer products); matrix Bernstein gives  $\|\hat{\Sigma} - \Sigma\| \leq \sqrt{\frac{\|\Sigma\| r \log d}{n}}$  ( $r$  = intrinsic dimension). [ fill in: fill the  $L, \nu$  estimates of the outer products and the sample-size threshold ]

Why it goes this way: the entire structure of scalar Bernstein (the variance term  $\nu$  + the uniform bound  $L$ , the two-regime  $\min$ ) survives untouched, with only an extra  $d$  (or intrinsic-dim) prefactor, coming from turning "one tail" into "the tail of  $d$  eigenvalues". This machine replaces the "union bound over an  $\epsilon$ -net" common in ML proofs, and is both tighter and less work.

This thread feeds directly into sketching and leverage-score sampling. [ [subgaussian](#)  $\leftrightarrow$  ]

 matrixbern

Martingale version: matrix Freedman (Tropp 2011) extends this to matrix martingales, pairing with §3. Needed for the spectral concentration of online / adaptive sampling.

## ■ 8 SUPREMA AND CHAINING

□□□□ The exit. When the target is  $\sup_t X_t$  (an empirical process), single-point Chernoff + a union bound is too loose.

In one line: the first seven sections control *one* random quantity. But ML/statistics really wants  $\sup_{t \in T} X_t$  (empirical processes, the sup in a generalization bound). **Chaining** uses multi-scale nets to control the supremum by an integral of metric entropy, reducing "infinitely many tails" to "the geometry of covering numbers". This is the bridge to empirical processes.

### The paradigm

**Trigger.** You want  $\mathbb{E} \sup_{t \in T} X_t$ , where  $\{X_t\}$  is a (sub-)Gaussian process with natural metric  $d(s, t) = \|X_s - X_t\|_{\psi_2}$ .

**Chaining:** along a sequence of refining  $\epsilon$ -nets  $T_0 \subset T_1 \subset \dots$ , telescope  $X_t - X_{t_0}$  and sum scale by scale. **Dudley:**

$$\mathbb{E} \sup_{t \in T} X_t \leq C \int_0^\infty \sqrt{\log N(T, d, \epsilon)} d\epsilon.$$

Three-line heuristic: (1) a union bound = a single scale, chaining = *all scales* summed, hence tighter; (2) the currency is **metric entropy**  $\log N(T, d, \epsilon)$ , turning a probability problem into covering geometry; (3) Dudley can be loose; the **generic chaining** quantity  $\gamma_2(T, d)$  is the *tight* one (Talagrand's majorizing measures).

### Cheat-sheet: from union bound to majorizing measures

| Tool              | Gives                                                       | When to use / tight?                                            |
|-------------------|-------------------------------------------------------------|-----------------------------------------------------------------|
| Union bound       | $\sqrt{2 \log  T }$ (finite $T$ )                           | opening move; too loose for large or continuous $T$             |
| Dudley integral   | $\int \sqrt{\log N(\varepsilon)} d\varepsilon$              | standard upper bound for continuous index; often enough         |
| Sudakov lower bd. | $\sup_{\varepsilon} \varepsilon \sqrt{\log N(\varepsilon)}$ | pairs with Dudley to pin the order                              |
| Generic chaining  | $\gamma_2(T, d)$                                            | matching upper <i>and</i> lower (Gaussian processes, Talagrand) |

## Main results & proof skeletons

**Theorem 8.1** (*Dudley's inequality*).  $\{X_t\}_{t \in T}$  a centered process, sub-Gaussian w.r.t.  $d$ . Then  $\mathbb{E} \sup_t X_t \leq C \int_0^\infty \sqrt{\log N(T, d, \varepsilon)} d\varepsilon$ .

*Proof.* Skeleton: take nets  $T_k$  at scale  $\varepsilon_k = 2^{-k}$  with nearest-point maps  $\pi_k(t)$ ; telescope  $X_t - X_{\pi_0(t)} = \sum_{k \geq 1} (X_{\pi_k(t)} - X_{\pi_{k-1}(t)})$ ; each link has  $|T_k| |T_{k-1}|$  sub-Gaussian increments, combine the per-scale union bound with the increment scale  $\lesssim 2^{-k}$  and sum. [ fill in: fill the per-scale union bound  $\sqrt{\log |T_k|}$  and the geometric-series sum  $\Rightarrow$  the integral ]

**Theorem 8.2** (*Generic chaining / majorizing measures*). For a Gaussian process,  $\mathbb{E} \sup_t X_t \asymp \gamma_2(T, d)$ , where  $\gamma_2(T, d) = \inf_{(T_k)} \sup_t \sum_{k \geq 0} 2^{k/2} d(t, T_k)$  with  $|T_k| \leq 2^{2^k}$ .

*Proof.* [ fill in: upper bound: weighted chaining (weights adapt to  $t$ , beating Dudley's uniform scales); lower bound: Talagrand's partition tree / iterated Sudakov. Cite the majorizing-measures theorem (do not reprove) ]

**Example 8.3** (*Operator norm via  $\varepsilon$ -net + chaining*).  $\|A\| = \sup_{u, v \in S^{n-1}} u^\top A v$ : discretize on a  $1/2$ -net of the sphere ( $\leq 9^n$  points) + each point sub-Gaussian (§2)  $\Rightarrow \mathbb{E} \|A\| \lesssim \sqrt{n}$ . Chaining “automates” the net argument to a continuous index. [ fill in: fill the standard  $1/2$ -net lemma lifting sup to  $2 \max_{\text{net}}$  ]

Why it goes this way: the Borell–TIS of §5 ([borelltis  $\leftrightarrow$ ]) controls the *fluctuation* of  $\sup_t X_t$ , this section its *expectation*, together they are the full concentration of an empirical process. This closes the loop with the “expectation = Rademacher complexity” half of the generalization bound in §3.

Chaining  $\rightarrow$  Rademacher complexity  $\rightarrow$  uniform laws of large numbers.

## OPEN THREADS · WHERE THIS GOES NEXT

□□□□

These notes are an **arsenal**, not a museum. The cross-section open questions below are the seeds for the next pass: a unifying perspective, the missing lower bounds, and the exit toward empirical processes.

~~~~~  
This is the exit:
metric entropy /
empirical
processes /
uniform laws
deserve *their*
own notebook
(Dudley, van der
Vaart, Wellner)

Open threads (cross-section)

- **Four faces:** concentration $\Leftrightarrow$ isoperimetry $\Leftrightarrow$ log-Sobolev $\Leftrightarrow$ transportation (Marton). Drawing the explicit commuting diagram of these equivalences is the “main theorem” this notebook most wants to add.
- **When does dimension enter?** Gaussian concentration is dimension-free, matrix concentration carries $\log d$, chaining carries metric entropy. A “sources of dimension dependence” table separates “truly dimension-free” from “hidden in the constants”.
- **Tightness:** is every upper bound paired with a lower bound? Cramér (§1) and Sudakov (§8) are; the lower bounds for McDiarmid and matrix Bernstein (worst f / worst spectrum) are to be filled.
- **The exit:** after §8, open a dedicated *empirical-process* notebook (VC / Rademacher / uniform laws / Donsker).

Closing line: the whole subject is seven variations on one move, *exponentiate the deviation, then crush it with structure*. What the structure is decides which engine you reach for. Recognizing the structure beats memorizing the formula.